

**HANOI UNIVERSITY OF SCIENCE AND
TECHNOLOGY**

**SCHOOL OF ELECTRICAL AND ELECTRONIC
ENGINEERING**

Project Report of Background Digital Communication

**Performance Evaluation of 2x2 MIMO-OFDM System
with Alamouti STBC and 16-QAM over Rayleigh Fading
Channels**

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Abstract

The objective of this project is to simulate and evaluate the performance of a 2x2 Multiple-Input Multiple-Output Orthogonal Frequency Division Multiplexing (MIMO-OFDM) communication system. The system utilizes 16-QAM modulation and Alamouti Space-Time Block Coding (STBC) over a multipath Rayleigh fading channel with Additive White Gaussian Noise (AWGN). Perfect Channel State Information (CSI) is assumed at the receiver to perform Maximum Likelihood detection. The system's performance is quantitatively evaluated using Bit Error Rate (BER) and Symbol Error Rate (SER) as functions of the Signal-to-Noise Ratio (SNR). Simulation results demonstrate the effectiveness of combining OFDM with Alamouti STBC to combat multipath fading and achieve significant diversity gain.

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1 Introduction

The rapid evolution of wireless communication demands robust systems capable of high data rates and reliable transmission over hostile propagation environments. This project simulates a 2x2 MIMO-OFDM communication system using 16-Quadrature Amplitude Modulation (16-QAM) over a multipath Rayleigh fading channel with Additive White Gaussian Noise (AWGN). Alamouti Space-Time Block Coding (STBC) is employed to provide transmit diversity. The system performance is evaluated by analyzing the Bit Error Rate (BER) and Symbol Error Rate (SER) across various Signal-to-Noise Ratio (SNR) levels.

2 Overview of Technologies Used

This project integrates several fundamental technologies in modern digital communication systems:

- **OFDM (Orthogonal Frequency Division Multiplexing):** Adopted for its high spectral efficiency and robustness against frequency-selective fading. By dividing the bandwidth into orthogonal subcarriers and employing a Cyclic Prefix (CP), OFDM mitigates Inter-Symbol Interference (ISI).
- **16-QAM (Quadrature Amplitude Modulation):** A high-order modulation scheme mapping 4 bits per symbol, significantly improving spectral efficiency compared to lower-order schemes.
- **MIMO and Alamouti STBC:** Multiple-Input Multiple-Output (MIMO) technology using the Alamouti STBC scheme to exploit spatial diversity, combat deep fading, and improve link reliability without requiring extra bandwidth.
- **Rayleigh Fading Channel:** Characterizes environments with rich multipath propagation. Additive White Gaussian Noise (AWGN) is also incorporated to model thermal noise at the receiver.

3 Theoretical Background

3.1 Orthogonal Frequency Division Multiplexing (OFDM)

Orthogonal Frequency-Division Multiplexing (OFDM) is a special case of FDM (Frequency Division Multiplexing). In the FDM technique, the total bandwidth of the

transmission link is divided into N non-overlapping frequency channels. The signal of each channel is modulated with a subcarrier, and the N channels are multiplexed using frequency division. This results in bandwidth inefficiency; to overcome this drawback of FDM, the OFDM technique was introduced.

This technique divides the frequency band into numerous sub-bands with different carriers, each of which is modulated to transmit a low-rate data stream. The aggregate of these low-rate data streams constitutes the high-rate data stream required for transmission. Furthermore, the carriers employed are orthogonal to each other, allowing their spectra to overlap without causing interference. Consequently, bandwidth utilization becomes significantly more efficient.

3.1.1 Orthogonality Principle

The fundamental concept of OFDM relies on the mathematical orthogonality of its subcarriers. Over a single symbol period T_s , two subcarriers with frequencies f_k and f_m are orthogonal if the integral of their product over the symbol duration equals zero. Mathematically, this continuous-time condition is expressed as:

$$\frac{1}{T_s} \int_0^{T_s} e^{j2\pi f_k t} \times e^{-j2\pi f_m t} dt = \begin{cases} 1, & \text{if } k = m \\ 0, & \text{if } k \neq m \end{cases} \quad (1)$$

To satisfy this condition, the subcarrier spacing Δf is chosen to be the reciprocal of the useful symbol duration $1/T_s$, such that $\Delta f = 1/T_s$. In the frequency domain, the spectrum of each subcarrier follows a sinc function shape. This precise spacing ensures that the peak of one subcarrier aligns with the nulls of all other subcarriers in the frequency domain. Consequently, crosstalk between subcarriers - known as Inter-Carrier Interference (ICI) is theoretically eliminated at the receiver's sampling instants.

3.1.2 Principles of using IFFT and FFT

In practical systems, using thousands of independent oscillators to generate orthogonal subcarriers is hardware-infeasible. Therefore, the Fourier transform is used as an effective alternative.

At the transmitter (using IFFT): Complex data symbols are mapped onto subcarriers in the frequency domain. The inverse fast Fourier transform (IFFT) is performed to convert these symbols to the time domain before transmission. The general formula for

the discrete signal $x(n)$ after IFFT is:

$$x(n) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi kn}{N}} \quad (2)$$

where $n = 0, 1, \dots, N - 1$ is the time-domain sample index. The scaling factor $1/\sqrt{N}$ ensures energy normalization.

At the receiver (using FFT): After receiving the time-domain signal, the receiver uses the Fast Fourier Transform (FFT) to convert the signal back to the frequency domain, thereby restoring the original data symbols.

Advantages: IFFT/FFT architecture significantly reduces computational complexity and makes system implementation hardware-feasible at a low cost.

3.1.3 Cyclic Prefix (CP) and ISI Mitigation

Cyclic Prefix (CP) is a guard interval inserted before each symbol to ensure transmission quality in multipath environments. The CP is created by copying the last N_{cp} samples of an OFDM symbol (after the IFFT block) and prepending them to the beginning of that same symbol.

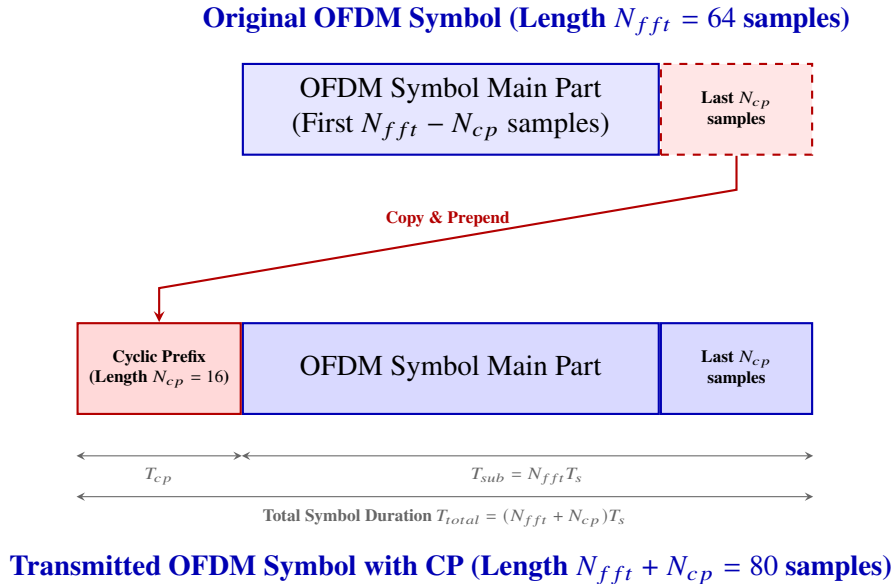


Figure 1: Cyclic Prefix (CP) Insertion Scheme in OFDM Symbol

Specifically, let the time-domain OFDM symbol after IFFT be represented as:

$$\mathbf{x} = [x(0), x(1), \dots, x(N - 1)] \quad (3)$$

The cyclic prefix consists of the last N_{cp} samples: $[x(N - N_{cp}), \dots, x(N - 1)]$. The transmitted symbol vector with CP becomes:

$$\mathbf{x}_{cp} = [x(N - N_{cp}), \dots, x(N - 1), x(0), x(1), \dots, x(N - 1)] \quad (4)$$

The role of the Cyclic Prefix can be detailed as follows:

- **Inter-Symbol Interference (ISI) Mitigation:** In a multipath propagation environment with a maximum channel delay spread of L taps, delayed replicas of the previous OFDM symbol overlap with the current symbol, causing ISI. If the CP length is chosen such that $N_{cp} \geq L - 1$, the delayed components of the previous symbol fall entirely within the CP guard interval and are discarded at the receiver, completely mitigating ISI.
- **Maintaining Orthogonality and Preventing ICI:** Without a CP, the propagation of the signal through a multipath channel corresponds to a linear convolution between the transmitted signal and the channel impulse response. This linear convolution breaks the orthogonality between subcarriers, leading to Inter-Carrier Interference (ICI). By prepending the CP, the linear convolution of the channel over the useful part of the symbol ($n = 0, \dots, N - 1$) is mathematically transformed into a circular convolution:

$$y(n) = x(n) \circledast h(n) + \eta(n), \quad 0 \leq n \leq N - 1 \quad (5)$$

where $\circledast$ denotes circular convolution, $h(n)$ is the channel impulse response, and $\eta(n)$ is the noise.

- **Receiver Simplification (One-Tap Equalization):** Due to the Discrete Fourier Transform (DFT) properties, circular convolution in the time domain is equivalent to point-wise multiplication in the frequency domain. Therefore, after taking the FFT at the receiver, the received signal at subcarrier k is given by:

$$Y(k) = H(k)X(k) + N(k) \quad (6)$$

where $H(k) = \text{FFT}(h)$ is the channel frequency response, and $N(k)$ is the frequency-domain noise. This allows the receiver to perform equalization using a simple division (one-tap equalization) per subcarrier:

$$\hat{X}(k) = \frac{Y(k)}{H(k)} \quad (7)$$

which avoids the need for computationally heavy time-domain equalizers.

3.2 Quadrature Amplitude Modulation (16-QAM)

3.2.1 Theory of 16-QAM Modulation

Quadrature Amplitude Modulation (QAM) is a digital modulation technique that transmits information by varying both the amplitude and phase of a carrier signal. The transmitted signal consists of two orthogonal components: In-phase component (I) and Quadrature component (Q).

For a 16-QAM system, the modulation order is $M = 16$. The number of bits carried by each symbol is $k = \log_2(M) = 4$ bits/symbol. Therefore, each transmitted symbol carries 4 bits of information.

The passband signal is given by:

$$s(t) = I \cos(2\pi f_c t) - Q \sin(2\pi f_c t) \quad (8)$$

where I is the In-phase component, Q is the Quadrature component, and f_c is the Carrier frequency.

The amplitude levels used in both I and Q branches are $I, Q \in \{-3, -1, +1, +3\}$. Since each axis contains four amplitude levels, the total number of constellation points is $N = 4 \times 4 = 16$. Thus, a 16-point constellation is obtained.

3.2.2 Average Symbol Energy Normalization

The average energy of the in-phase component is calculated as:

$$\mathbb{E}[I^2] = \frac{(-3)^2 + (-1)^2 + (+1)^2 + (+3)^2}{4} = \frac{9 + 1 + 1 + 9}{4} = 5 \quad (9)$$

Similarly, $\mathbb{E}[Q^2] = 5$. Therefore, the average symbol energy becomes:

$$E_s = \mathbb{E}[I^2] + \mathbb{E}[Q^2] = 5 + 5 = 10 \quad (10)$$

To normalize the constellation such that $E_s = 1$, all constellation points are multiplied by $1/\sqrt{10}$. Hence, the normalized transmitted symbol is:

$$s = \frac{I + jQ}{\sqrt{10}} \quad (11)$$

This normalization is used in the MATLAB implementation and simplifies BER and SER calculations.

3.2.3 16-QAM Constellation Diagram and Gray Bit Mapping

Gray coding is employed so that adjacent constellation points differ by only one bit. The first two bits determine the I component, while the last two bits determine the Q component.

Table 1: Constellation Coordinates and Gray Mapping

Q \ I	I = -3 (00)	I = -1 (01)	I = +1 (11)	I = +3 (10)
Q = +3 (10)	(-3, 3) / 0010	(-1, 3) / 0110	(1, 3) / 1110	(3, 3) / 1010
Q = +1 (11)	(-3, 1) / 0011	(-1, 1) / 0111	(1, 1) / 1111	(3, 1) / 1011
Q = -1 (01)	(-3, -1) / 0001	(-1, -1) / 0101	(1, -1) / 1101	(3, -1) / 1001
Q = -3 (00)	(-3, -3) / 0000	(-1, -3) / 0100	(1, -3) / 1100	(3, -3) / 1000

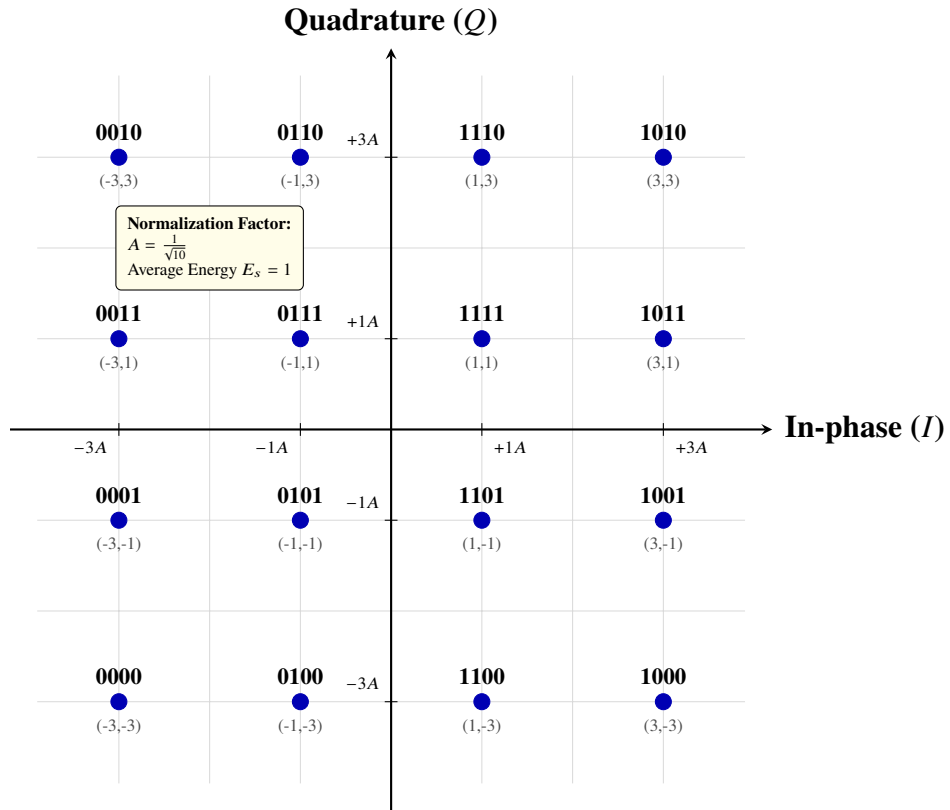


Figure 2: 16-QAM Constellation Diagram with Gray Mapping

To clarify the mapping rule, several examples are presented:

- **Example 1 (Input bits 0000):** The first two bits 00 correspond to $I = -3$, and the last two bits 00 correspond to $Q = -3$. The resulting complex symbol is $s = \frac{-3-j3}{\sqrt{10}}$.
- **Example 2 (Input bits 1111):** The first two bits 11 correspond to $I = +1$, and the last two bits 11 correspond to $Q = +1$. The resulting complex symbol is $s = \frac{1+j1}{\sqrt{10}}$.
- **Example 3 (Input bits 1010):** The first two bits 10 correspond to $I = +3$, and the last two bits 10 correspond to $Q = +3$. The resulting complex symbol is $s = \frac{3+j3}{\sqrt{10}}$.

3.2.4 Euclidean Distance Demodulation

After transmission through the Rayleigh fading channel and AWGN, the received symbol is represented as $r = I_r + jQ_r$. Let the m -th constellation point be $S_m = I_m + jQ_m$. The Euclidean distance between the received symbol and the constellation point is:

$$d_m = \sqrt{(I_r - I_m)^2 + (Q_r - Q_m)^2} \quad (12)$$

To reduce computational complexity, the receiver usually employs the squared Euclidean distance. The Maximum Likelihood (ML) decision rule is:

$$\hat{S} = \arg \min_{S_m} \left((I_r - I_m)^2 + (Q_r - Q_m)^2 \right) \quad (13)$$

The constellation point with the minimum Euclidean distance is selected as the detected symbol and then converted back into the corresponding 4-bit sequence.

3.3 MIMO (Multiple Input Multiple Output) and Alamouti STBC

MIMO is a key technique in modern wireless communication systems that utilizes multiple antennas at both the transmitter and receiver. By transmitting and receiving multiple signals simultaneously over the same frequency band, MIMO systems exploit the spatial dimension to achieve:

- **Spatial Multiplexing Gain:** Transmitting independent data streams from different antennas to linearly scale the system capacity with the number of antennas.
- **Diversity Gain:** Transmitting the same or related signals through multiple independent fading paths to improve transmission reliability and combat channel fading.

A 2×2 MIMO system consists of $N_T = 2$ transmit antennas and $N_R = 2$ receive antennas, creating four spatial communication links. In a flat-fading or subcarrier-level

frequency-domain representation, the received signal vector $\mathbf{y} = [y_1, y_2]^T$ is modeled as:

$$\mathbf{y} = \mathbf{H}\mathbf{s} + \boldsymbol{\eta} \quad (14)$$

where $\mathbf{s} = [s_1, s_2]^T$ is the transmitted symbol vector, $\boldsymbol{\eta} = [\eta_1, \eta_2]^T$ is the additive white Gaussian noise (AWGN) vector, and $\mathbf{H}$ is the 2×2 channel matrix:

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \quad (15)$$

Here, h_{ij} represents the complex channel path gain from transmit antenna j to receive antenna i .

3.3.1 MIMO-OFDM Integration

When MIMO is combined with OFDM, the wideband frequency-selective MIMO channel is divided into N_{fft} parallel, orthogonal narrowband flat-fading MIMO subchannels. For each subcarrier $k \in \{0, 1, \dots, N_{fft} - 1\}$, the relationship between the transmitted symbol vector $\mathbf{s}(k) = [s_1(k), s_2(k)]^T$ and the received vector $\mathbf{y}(k) = [y_1(k), y_2(k)]^T$ is given by:

$$\mathbf{y}(k) = \mathbf{H}(k)\mathbf{s}(k) + \boldsymbol{\eta}(k) \quad (16)$$

where $\mathbf{H}(k)$ is the estimated channel response matrix at subcarrier k , containing the channel gains $h_{ij}(k)$ between transmit antenna j and receive antenna i at frequency subcarrier k .

3.3.2 Practical Transmitter Components: Channel Coding and Interleaving

In practical MIMO-OFDM transmitters (such as in IEEE 802.11 or LTE), the input binary stream is processed through several additional stages before being mapped into symbols:

- **Channel Encoder:** Adds redundant bits (using convolutional codes, Turbo codes, or LDPC codes) to enable forward error correction (FEC) at the receiver.
- **Interleaver (IL):** Permutes the coded bits in a pseudo-random order. This is crucial in OFDM systems since deep fading on certain subcarriers can corrupt consecutive bits. Interleaving spreads these errors across the entire block, transforming burst errors into random, single-bit errors that the channel decoder can easily correct.

While these blocks are essential for commercial systems to achieve the required Quality of Service (QoS), they are omitted in this simulation to focus strictly on evaluating the

raw physical layer performance of the 16-QAM MIMO-OFDM scheme with Alamouti STBC under fading channels.

3.3.3 Alamouti STBC Encoding

To exploit diversity gain without needing multiple RF chains or channel estimation at the transmitter, the Alamouti Space-Time Block Coding (STBC) scheme is employed. The code operates across two consecutive OFDM symbols (or time slots) $t = 1, 2$.

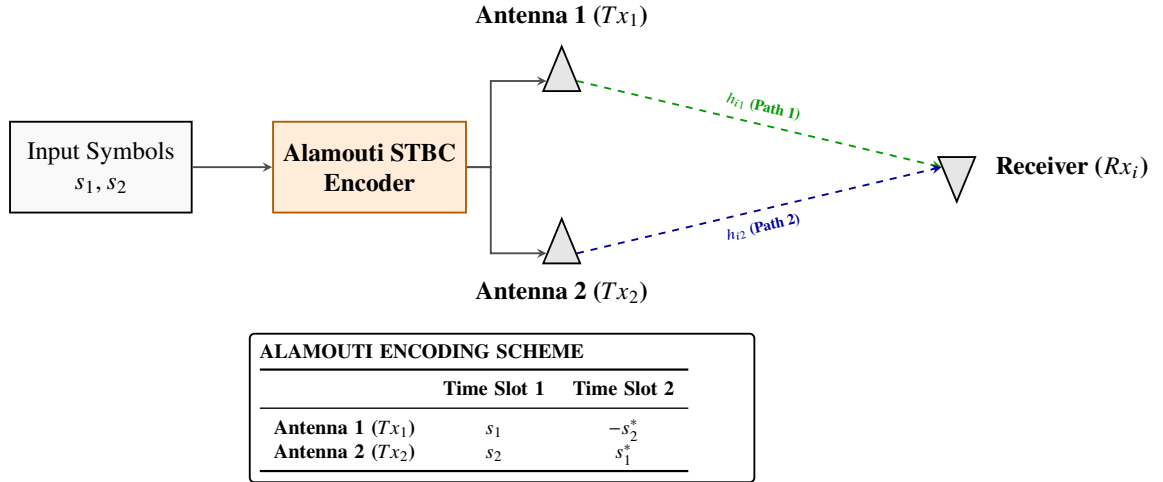


Figure 3: Alamouti Space-Time Block Coding Transmission Scheme

The implemented system uses a 2x2 Alamouti Space-Time Block Code (STBC). For two consecutive symbols s_1 and s_2 , the transmission matrix over two time slots is:

$$\begin{bmatrix} s_1 & s_2 \\ -s_2^* & s_1^* \end{bmatrix} \quad (17)$$

where $(*)$ denotes complex conjugation. This coding scheme provides transmit diversity and improves performance in fading environments.

3.3.4 Signal Model at Receiver

In the first time slot, Antenna 1 transmits $s_1(k)$ and Antenna 2 transmits $s_2(k)$ on subcarrier k . The received signals at receive antenna $i \in \{1, 2\}$ in time slot 1 are:

$$y_{i1}(k) = h_{i1}(k)s_1(k) + h_{i2}(k)s_2(k) + \eta_{i1}(k) \quad (18)$$

which can be written in matrix form as:

$$\begin{bmatrix} y_{11}(k) \\ y_{21}(k) \end{bmatrix} = \begin{bmatrix} h_{11}(k) & h_{12}(k) \\ h_{21}(k) & h_{22}(k) \end{bmatrix} \begin{bmatrix} s_1(k) \\ s_2(k) \end{bmatrix} + \begin{bmatrix} \eta_{11}(k) \\ \eta_{21}(k) \end{bmatrix} \quad (19)$$

In the second time slot, Antenna 1 transmits $-s_2^*(k)$ and Antenna 2 transmits $s_1^*(k)$ on subcarrier k . The received signals at receive antenna $i \in \{1, 2\}$ in time slot 2 are:

$$y_{i2}(k) = -h_{i1}(k)s_2^*(k) + h_{i2}(k)s_1^*(k) + \eta_{i2}(k) \quad (20)$$

which in matrix form is:

$$\begin{bmatrix} y_{12}(k) \\ y_{22}(k) \end{bmatrix} = \begin{bmatrix} h_{11}(k) & h_{12}(k) \\ h_{21}(k) & h_{22}(k) \end{bmatrix} \begin{bmatrix} -s_2^*(k) \\ s_1^*(k) \end{bmatrix} + \begin{bmatrix} \eta_{12}(k) \\ \eta_{22}(k) \end{bmatrix} \quad (21)$$

3.3.5 Alamouti STBC Decoding

Assuming Perfect Channel State Information (CSI) at the receiver (meaning the gains $h_{ij}(k)$ are known), the transmitted symbols $s_1(k)$ and $s_2(k)$ are estimated by combining the received signals across both receive antennas and both time slots. The combination equations are:

$$\tilde{s}_1(k) = \sum_{i=1}^2 (h_{i1}^*(k)y_{i1}(k) + h_{i2}(k)y_{i2}^*(k)) \quad (22)$$

$$\tilde{s}_2(k) = \sum_{i=1}^2 (h_{i2}^*(k)y_{i1}(k) - h_{i1}(k)y_{i2}^*(k)) \quad (23)$$

To compute the final estimates for Maximum Likelihood (ML) detection, these combined signals are normalized by the total channel power of the active links:

$$\hat{s}_1(k) = \frac{\tilde{s}_1(k)}{\sum_{i=1}^2 (|h_{i1}(k)|^2 + |h_{i2}(k)|^2)} \quad (24)$$

$$\hat{s}_2(k) = \frac{\tilde{s}_2(k)}{\sum_{i=1}^2 (|h_{i1}(k)|^2 + |h_{i2}(k)|^2)} \quad (25)$$

These decision variables $\hat{s}_1(k)$ and $\hat{s}_2(k)$ are then passed to the Euclidean distance detector to find the closest constellation point in the complex plane.

4 System Block Diagram and Operation

4.1 System Block Diagram

To understand the interaction between the components, the complete block diagram of the simulated 2×2 MIMO-OFDM communication system is shown below.

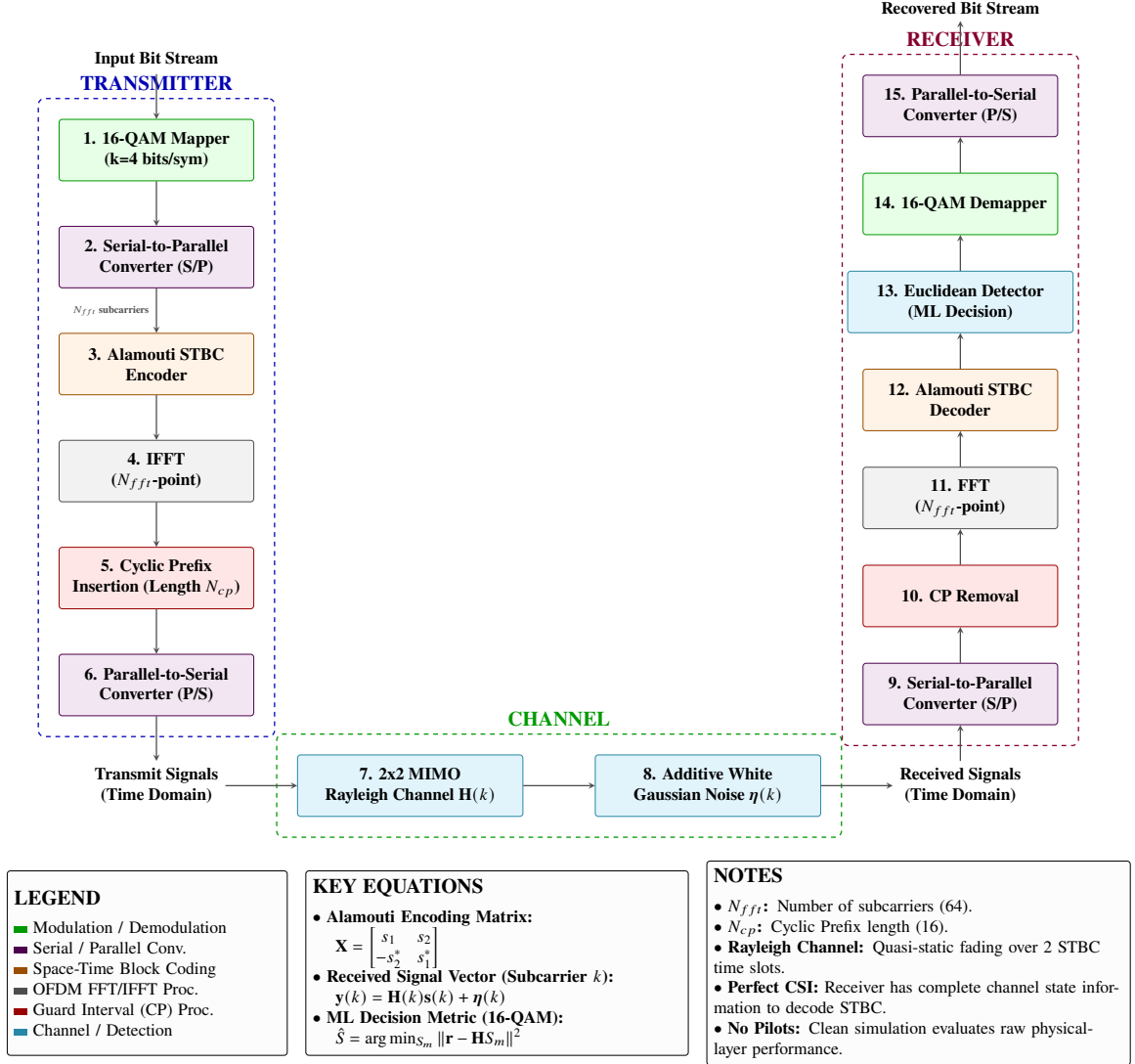


Figure 4: Detailed System Block Diagram of the 2×2 MIMO-OFDM System

The processing chain is structured into three main segments:

- **Transmitter:**

1. *Bit Generator*: Generates a stream of pseudo-random binary bits.
2. *Serial to Parallel (S/P) Converter*: Splits the serial bit stream into parallel blocks to feed the mapper.
3. *16-QAM Mapper*: Groups the bits into 4-bit symbols and maps them onto the complex 16-QAM constellation.
4. *Alamouti STBC Encoder*: Encodes the mapped symbols in pairs (s_1, s_2) across two spatial paths and two time slots.
5. *OFDM Modulator (IFFT + CP)*: Applies an N_{fft} -point Inverse Fast Fourier Transform (IFFT) to convert frequency subcarriers to time-domain samples, and prepends a Cyclic Prefix (CP) of length N_{cp} to mitigate multipath interference.

- **Transmission Channel:**

1. *Multipath Rayleigh Fading*: Simulates a multipath channel environment using a 6-tap exponentially decaying power delay profile. The fading is assumed to be quasi-static (constant over two Alamouti symbol periods).
2. *Additive White Gaussian Noise (AWGN)*: Models thermal noise at each receive antenna.

- **Receiver:**

1. *OFDM Demodulator (Remove CP + FFT)*: Discards the CP and applies an N_{fft} -point Fast Fourier Transform (FFT) to return the signals to the frequency domain.
2. *Alamouti STBC Decoder*: Combines received signals across the two time slots and two antennas using channel estimates (Perfect CSI) to separate and estimate the transmitted symbols.
3. *Euclidean Detector*: Computes the squared Euclidean distance between the estimated symbols and the normalized 16-QAM constellation points.
4. *16-QAM Demapper*: Recovers the 4-bit binary sequences from the detected symbols.
5. *Parallel to Serial (P/S) Converter*: Restores the single serial bit stream for final performance evaluation.

4.2 System Operation

1. **Bit Generation:** A random binary bit stream is generated.
2. **16-QAM Mapping:** Every four bits are mapped into one normalized 16-QAM symbol according to the Gray coding rule.
3. **Alamouti STBC Encoding:** The symbols are encoded using Alamouti STBC to exploit spatial diversity.
4. **OFDM Modulation:** The encoded symbols are transformed into the time domain using an Inverse Fast Fourier Transform (IFFT). A Cyclic Prefix (CP) is added to mitigate inter-symbol interference.
5. **Channel Transmission:** The OFDM signal propagates through a multipath Rayleigh fading channel and is corrupted by Additive White Gaussian Noise (AWGN).
6. **OFDM Demodulation:** At the receiver, the cyclic prefix is removed and a Fast Fourier Transform (FFT) is performed.
7. **Alamouti STBC Decoding:** The receiver combines signals from multiple antennas and estimates the transmitted symbols using Perfect CSI.
8. **Euclidean Distance Detection:** The Euclidean distance between the received symbol and all possible constellation points is calculated. The constellation point with the minimum distance is selected.
9. **16-QAM Demapping:** The detected symbol is converted back into the corresponding binary sequence using the Gray mapping rule.
10. **Performance Evaluation:** The recovered bit stream is compared with the transmitted bit stream to evaluate BER and SER performance.

5 Experimental Setup

The simulation framework was developed using a MATLAB-based environment to evaluate the performance of a 2x2 MIMO-OFDM system. The proposed system models the transmission of a random bit stream through a multipath Rayleigh fading channel with Additive White Gaussian Noise (AWGN).

5.1 Simulation Parameters

To ensure reproducibility and clarity, the main parameters used in the simulation are summarized in Table 2.

Table 2: Simulation Parameters

Parameter	Value
Number of Transmit Antennas	2
Number of Receive Antennas	2
MIMO Scheme	Alamouti STBC
FFT Size (N)	64
Cyclic Prefix (CP) Length	16
Modulation Scheme	16-QAM
Bits per Symbol	4
Channel Type	Multipath Rayleigh Fading
Number of Channel Taps	6
Noise Model	AWGN
E_b/N_0 Range	0 dB to 30 dB
Performance Metrics	BER and SER
Channel State Information	Perfect CSI assumed at receiver

5.2 Evaluation Scenario

The simulation is repeated for E_b/N_0 values from 0 dB to 30 dB. For each value, a large number of random bits are transmitted through the 2x2 MIMO-OFDM system. The corresponding BER and SER values are collected and plotted on a logarithmic scale. These curves are then used to evaluate how the system performance changes as the signal-to-noise ratio increases. The expected result is that both BER and SER decrease as E_b/N_0 increases. The use of Alamouti STBC is expected to improve system reliability by providing spatial diversity, while OFDM helps reduce the effect of multipath propagation by converting the frequency-selective channel into multiple narrowband subchannels.

6 Results and Discussion

The performance of the 2x2 MIMO-OFDM system utilizing Alamouti Space-Time Block Coding (STBC) and 16-QAM modulation over a Rayleigh fading channel with additive white Gaussian noise (AWGN) is evaluated by measuring the Bit Error Rate (BER) and Symbol Error Rate (SER) across various Signal-to-Noise Ratio (SNR) levels.

6.1 Simulation Results

The obtained simulation results illustrate the variations of BER and SER as functions of E_b/N_0 . The corresponding performance curves are plotted in Figure 5. To provide quantitative precision, specific performance data points extracted from the simulation with a step size of 2 dB are presented in Table 3.

Table 3: BER and SER Performance Data of the 16-QAM MIMO-OFDM System

E_b/N_0 (dB)	Bit Error Rate (BER)	Symbol Error Rate (SER)
0	9.11283e-02	3.19521e-01
2	5.59388e-02	2.03760e-01
4	2.97852e-02	1.11914e-01
6	1.36120e-02	5.18828e-02
8	4.64779e-03	1.80755e-02
10	1.39518e-03	5.45573e-03
12	4.33594e-04	1.69792e-03
14	7.42187e-05	2.96875e-04
16	1.75781e-05	6.77083e-05
18	6.51042e-07	2.60417e-06
20	6.51042e-07	2.60417e-06
22	6.51042e-07	2.60417e-06
24	0.00000e+00	0.00000e+00
26	0.00000e+00	0.00000e+00
28	0.00000e+00	0.00000e+00
30	0.00000e+00	0.00000e+00

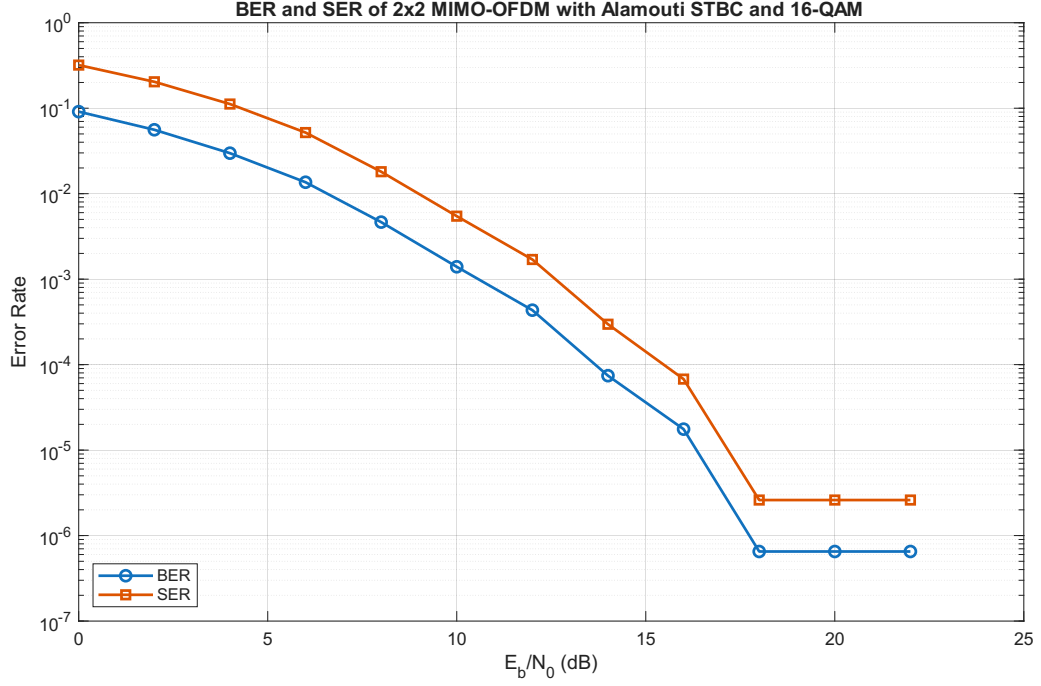


Figure 5: BER and SER Performance of the 2x2 MIMO-OFDM System with Alamouti STBC and 16-QAM

6.2 Discussion

6.2.1 General Performance Trend

The simulation results show that both BER and SER decrease as E_b/N_0 increases. At low E_b/N_0 , the received signal is strongly affected by Rayleigh fading and AWGN, resulting in relatively high error rates. As E_b/N_0 increases, the noise effect becomes weaker, so the detection accuracy improves and the BER/SER values decrease significantly.

6.2.2 BER and SER Comparison

The SER values are higher than the BER values because each 16-QAM symbol carries 4 bits. A symbol error occurs when at least one bit in the symbol is detected incorrectly. However, the use of Gray coding helps reduce the number of bit errors when the received symbol is mistaken for a neighboring constellation point.

6.2.3 Diversity Gain from Alamouti STBC

The use of Alamouti STBC in the 2x2 MIMO system provides spatial diversity. By transmitting coded symbols through two transmit antennas and combining the received

signals from two receive antennas, the system reduces the impact of deep fading. This explains the rapid decrease of BER and SER as E_b/N_0 increases.

6.2.4 Effect of OFDM in Multipath Channel

OFDM improves system performance in the multipath Rayleigh channel. The IFFT/FFT structure divides the frequency-selective channel into multiple narrowband subchannels, while the cyclic prefix helps reduce inter-symbol interference. In this simulation, the cyclic prefix length is sufficient for the 6-tap channel, so most ISI effects are mitigated.

6.2.5 Simulation Resolution Limit in the High- E_b/N_0 Region

In the high- E_b/N_0 region (specifically from 18 dB to 22 dB), a flattening of the BER and SER curves can be observed at exactly 6.51×10^{-7} and 2.60×10^{-6} , respectively. As the simulation assumes perfect channel state information (CSI) at the receiver, this is not a physical error floor. Rather, it represents the **resolution limit of the Monte Carlo simulation** due to the finite number of transmitted bits.

With the configured $N_{blocks} = 3000$, the total number of transmitted bits per SNR point is 1,536,000. At these high SNR levels, exactly **one bit error** (and correspondingly one symbol error) occurs across the entire simulation, causing the calculated error rates to bottleneck at a constant minimum measurable value ($1/1,536,000 \approx 6.51 \times 10^{-7}$). Once E_b/N_0 reaches 24 dB, the system becomes entirely error-free (0 errors), causing the BER and SER to drop to strictly zero. To accurately capture and observe a smoother waterfall curve for error probabilities below this threshold, the number of simulation blocks must be significantly increased (e.g., $N_{blocks} \geq 10^5$).

6.2.6 Overall Evaluation

Overall, the results confirm that the 2x2 MIMO-OFDM system using Alamouti STBC and 16-QAM achieves good performance over a multipath Rayleigh fading channel. The performance improvement mainly comes from the diversity gain of STBC and the ability of OFDM to handle multipath propagation.

7 Conclusion

This project presented the simulation and performance evaluation of a 2x2 MIMO-OFDM communication system using 16-QAM modulation and Alamouti Space-Time Block Coding (STBC) over a multipath Rayleigh fading channel with Additive White

Gaussian Noise (AWGN). Perfect Channel State Information (CSI) was assumed at the receiver to perform Maximum Likelihood detection.

The simulation results demonstrated that both Bit Error Rate (BER) and Symbol Error Rate (SER) decrease as the Signal-to-Noise Ratio (SNR) increases. The implemented architecture improves spectral efficiency, provides spatial diversity gain, and enhances communication reliability in wireless fading environments. The use of OFDM effectively handles multipath propagation by eliminating inter-symbol interference through the use of a cyclic prefix. Overall, the project provided practical insights into the behavior of modern wireless communication systems.

8 References

1. J. G. Proakis, *Digital Communications*, 5th ed., McGraw-Hill, 2008.
2. S. Haykin, *Digital Communication Systems*, Wiley, 2013.
3. T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed., Prentice Hall, 2002.

A MATLAB Simulation Source Code

The following complete and self-contained MATLAB script was used to perform the Monte Carlo simulation of the 2x2 MIMO-OFDM system with Alamouti STBC and 16-QAM modulation over Rayleigh fading channels:

```
1 clc;
2 clear;
3 close all;
4
5 %% =====
6 % 2x2 MIMO-OFDM with Alamouti STBC over multipath Rayleigh channel
7 % Modulation: 16-QAM
8 % Metrics: BER and SER versus Eb/N0
9 % No Communications Toolbox required
10 % =====
11
12 rng(1); % For reproducible results
13
14 %% ----- Simulation parameters -----
15 Nt = 2; % Number of transmit antennas
16 Nr = 2; % Number of receive antennas
17
18 M = 16; % 16-QAM
19 k = log2(M); % Bits per QAM symbol = 4
20
21 Nfft = 64; % Number of OFDM subcarriers
22 Ncp = 16; % Cyclic prefix length
23 Lch = 6; % Number of Rayleigh multipath taps
24
25 EbN0_dB = 0:2:30; % Eb/N0 range in dB
26 Nblocks = 3000; % Number of Alamouti-OFDM blocks per
    Eb/N0
27 txScale = 1 / sqrt(Nt); % Normalize total transmit power
28
29 BER = zeros(size(EbN0_dB));
30 SER = zeros(size(EbN0_dB));
31
32 % Exponential power delay profile (PDP)
33 pdp = exp(-(0:Lch-1));
34 pdp = pdp / sum(pdp); % Normalize total channel power to 1
35
36 %% =====
```

```

37 %                               Main Eb/N0 simulation loop
38 % =====
39
40 for isnr = 1:length(EbN0_dB)
41     EbN0_linear = 10^(EbN0_dB(isnr)/10);
42     noiseVar = 1 / (k * EbN0_linear); % Noise variance per complex
sample
43
44     totalBitErrors = 0;
45     totalSymErrors = 0;
46     totalBits = 0;
47     totalSymbols = 0;
48
49     for blk = 1:Nblocks
50         % 1. Generate random bits
51         NsymPerBlock = 2 * Nfft;
52         NbitsPerBlock = NsymPerBlock * k;
53         txBits = randi([0 1], NbitsPerBlock, 1);
54
55         % 2. 16-QAM mapping
56         txSymbols = qam16_mod(txBits);
57         s1 = txSymbols(1:Nfft).';
58         s2 = txSymbols(Nfft+1:end).';
59
60         % 3. Alamouti STBC encoding in frequency domain
61         X_tx1_slot1 = txScale * s1;
62         X_tx2_slot1 = txScale * s2;
63         X_tx1_slot2 = txScale * (-conj(s2));
64         X_tx2_slot2 = txScale * ( conj(s1));
65
66         % 4. OFDM modulation: IFFT + cyclic prefix
67         x_tx1_slot1 = ofdm_mod(X_tx1_slot1, Nfft, Ncp);
68         x_tx2_slot1 = ofdm_mod(X_tx2_slot1, Nfft, Ncp);
69         x_tx1_slot2 = ofdm_mod(X_tx1_slot2, Nfft, Ncp);
70         x_tx2_slot2 = ofdm_mod(X_tx2_slot2, Nfft, Ncp);
71
72         % 5. Generate 2x2 multipath Rayleigh channel
73         h = zeros(Nr, Nt, Lch);
74         for rx = 1:Nr
75             for tx = 1:Nt
76                 h(rx, tx, :) = sqrt(pdp/2) .* (randn(1, Lch) + 1j*
randn(1, Lch));
77             end

```

```

78     end
79
80     % 6. Channel transmission + AWGN
81     y_slot1 = zeros(Nr, Nfft + Ncp + Lch - 1);
82     y_slot2 = zeros(Nr, Nfft + Ncp + Lch - 1);
83
84     for rx = 1:Nr
85         h_rx_tx1 = squeeze(h(rx, 1, :)).';
86         h_rx_tx2 = squeeze(h(rx, 2, :)).';
87
88         y_slot1(rx, :) = conv(x_tx1_slot1, h_rx_tx1) + conv(
x_tx2_slot1, h_rx_tx2);
89         y_slot2(rx, :) = conv(x_tx1_slot2, h_rx_tx1) + conv(
x_tx2_slot2, h_rx_tx2);
90     end
91
92     % Add AWGN
93     noise1 = sqrt(noiseVar/2) * (randn(size(y_slot1)) + 1j*randn(
size(y_slot1)));
94     noise2 = sqrt(noiseVar/2) * (randn(size(y_slot2)) + 1j*randn(
size(y_slot2)));
95     y_slot1 = y_slot1 + noise1;
96     y_slot2 = y_slot2 + noise2;
97
98     % 7. OFDM receiver: remove CP + FFT
99     Y1 = zeros(Nr, Nfft);
100    Y2 = zeros(Nr, Nfft);
101    for rx = 1:Nr
102        Y1(rx, :) = ofdm_demod(y_slot1(rx, :), Nfft, Ncp);
103        Y2(rx, :) = ofdm_demod(y_slot2(rx, :), Nfft, Ncp);
104    end
105
106    % 8. Channel frequency response
107    H = zeros(Nr, Nt, Nfft);
108    for rx = 1:Nr
109        for tx = 1:Nt
110            h_temp = squeeze(h(rx, tx, :)).';
111            H(rx, tx, :) = fft([h_temp zeros(1, Nfft - Lch)], Nfft
);
112        end
113    end
114    G = txScale * H;
115

```

```

116 % 9. Alamouti STBC decoding (2 receive antennas)
117 s1_hat = zeros(1, Nfft);
118 s2_hat = zeros(1, Nfft);
119
120 for sc = 1:Nfft
121     numerator_s1 = 0;
122     numerator_s2 = 0;
123     denominator = 0;
124
125     for rx = 1:Nr
126         g1 = G(rx, 1, sc);
127         g2 = G(rx, 2, sc);
128         y1 = Y1(rx, sc);
129         y2 = Y2(rx, sc);
130
131         numerator_s1 = numerator_s1 + conj(g1)*y1 + g2*conj(y2
132 );
133         numerator_s2 = numerator_s2 + conj(g2)*y1 - g1*conj(y2
134 );
135         denominator = denominator + abs(g1)^2 + abs(g2)^2;
136     end
137
138     s1_hat(sc) = numerator_s1 / denominator;
139     s2_hat(sc) = numerator_s2 / denominator;
140 end
141
142 rxSymbols = [s1_hat.'; s2_hat.'];
143
144 % 10. 16-QAM demodulation
145 rxBits = qam16_demod(rxSymbols);
146
147 % 11. Error metrics calculation
148 bitErrors = sum(txBits ~= rxBits);
149 txBitMatrix = reshape(txBits, k, []).';
150 rxBitMatrix = reshape(rxBits, k, []).';
151 symErrors = sum(any(txBitMatrix ~= rxBitMatrix, 2));
152
153 totalBitErrors = totalBitErrors + bitErrors;
154 totalSymErrors = totalSymErrors + symErrors;
155 totalBits = totalBits + NbitsPerBlock;
156 totalSymbols = totalSymbols + NsymPerBlock;
157 end

```

```

157     BER(isnr) = totalBitErrors / totalBits;
158     SER(isnr) = totalSymErrors / totalSymbols;
159
160     fprintf('Eb/N0 = %2d dB, BER = %.5e, SER = %.5e\n', ...
161         EbN0_dB(isnr), BER(isnr), SER(isnr));
162 end
163
164 %% =====
165 % 12. Plot BER and SER Results
166 % =====
167 figure;
168 semilogy(EbN0_dB, BER, 'o-', 'LineWidth', 1.5, 'MarkerFaceColor', 'b')
169     ;
170 hold on;
171 semilogy(EbN0_dB, SER, 's-', 'LineWidth', 1.5, 'MarkerFaceColor', 'r')
172     ;
173 grid on;
174 xlabel('E_b/N_0 (dB)');
175 ylabel('Error Probability');
176 title('Performance of 2x2 MIMO-OFDM with Alamouti STBC and 16-QAM');
177 legend('BER', 'SER', 'Location', 'southwest');
178
179 %% =====
180 % Local functions
181 % =====
182
183 function symbols = qam16_mod(bits)
184     bits = bits(:);
185     if mod(length(bits), 4) ~= 0
186         error('Number of bits must be a multiple of 4 for 16-QAM.');
```

```

198     Q = imag(symbols);
199
200     bitsI = gray_level_to_bits(I);
201     bitsQ = gray_level_to_bits(Q);
202     bitGroups = [bitsI bitsQ];
203     bits = reshape(bitGroups.', [], 1);
204 end
205
206 function levels = bits_to_gray_level(bitPairs)
207     % Gray mapping: 00 -> -3, 01 -> -1, 11 -> +1, 10 -> +3
208     levels = zeros(size(bitPairs, 1), 1);
209     for i = 1:size(bitPairs, 1)
210         b1 = bitPairs(i, 1);
211         b2 = bitPairs(i, 2);
212         if b1 == 0 && b2 == 0, levels(i) = -3;
213         elseif b1 == 0 && b2 == 1, levels(i) = -1;
214         elseif b1 == 1 && b2 == 1, levels(i) = 1;
215         elseif b1 == 1 && b2 == 0, levels(i) = 3;
216         end
217     end
218 end
219
220 function bitPairs = gray_level_to_bits(values)
221     % Gray demapping decision boundaries
222     bitPairs = zeros(length(values), 2);
223     for i = 1:length(values)
224         v = values(i);
225         if v < -2, bitPairs(i, :) = [0 0];
226         elseif v < 0, bitPairs(i, :) = [0 1];
227         elseif v < 2, bitPairs(i, :) = [1 1];
228         else, bitPairs(i, :) = [1 0];
229         end
230     end
231 end
232
233 function txTime = ofdm_mod(X, Nfft, Ncp)
234     x = ifft(X, Nfft) * sqrt(Nfft);
235     txTime = [x(end-Ncp+1:end) x];
236 end
237
238 function X = ofdm_demod(rxTime, Nfft, Ncp)
239     X = fft(rxTime(Ncp+1:Ncp+Nfft), Nfft) / sqrt(Nfft);
240 end

```